

Glass, Frames and Mechanisms



Special Tool(s)

 ST2574-A	Flex Probe Kit 105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST3094-A	Test Light SGT27000 or equivalent 250-300 mA incandescent bulb test lamp
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Delayed Accessory Power

The Body Control Module (BCM) controls voltage to the power windows with the accessory delay relay. The BCM activates the accessory delay relay whenever the ignition is in the RUN or the ACC position, or when the ignition is changed from RUN or ACC to the OFF position and the LH and RH front doors are closed.

The BCM deactivates the accessory delay relay when:

- the LH front door is ajar and the ignition is in the OFF or key-out position.
- the RH front door is ajar and the ignition is in the OFF or key-out position.
- ten minutes have elapsed since the ignition was changed from ACC or RUN to the OFF position.

LH and RH Front Power Window Control

The LH and RH front door window motors have integral electronics and must be initialized whenever a new window motor has been installed. Initialization is required to learn the full UP position and the profile of the glass as it travels through the glass channel. Once initialized, obstacle detection is enabled. When mechanical repairs have been carried out on the window regulator or glass run, the window motor must be de-initialized, and then initialized. The window control switch sends 2 separate signals to the window motor: up and down. When the window control switch is pressed to the first detent position, a 12-volt signal is sent to the window motor to request an up or down operation. When auto up or auto down is requested

(switch pressed to the second detent position), the window control switch provides a 12-volt signal on both the up and down line simultaneously. During an auto up or auto down request, the window motor determines intended window direction by the signal first received. If the window control switch is pressed too quickly to the second detent position (less than 5 milliseconds of time between first detent signal and second detent signal), the window motor is not able to determine the intended direction request and cannot operate until the window control switch is released and pressed again. The up and down contacts are floating within the window control switch when the switch is in the neutral position. The up and down feeds to the window motor are all low current.

The LH and RH front window motors operate when the delayed accessory relay is active. The high current required to move the front windows are supplied through the B+ and motor ground input.

The LH and RH front window regulator motors have a protection shutdown feature that disables the motor operation if the windows are continually operated up and down for a number of cycles (6-10). The feature disables the motor operation for 5 minutes to prevent damage to the regulator motors. Normal operation will resume when the window switch has been inactive for 5 minutes.

When the LH or RH front window motor is operating in auto up or auto down mode, movement of the front window can be stopped by pressing the switch to any position (UP, DOWN, auto UP, auto DOWN). The front window control switches must be released before the window can move again.

A new LH or RH front window motor cannot operate in one-touch up mode and the bounce-back feature is disabled prior to initialization. If the switch is actuated to the auto UP position and released, window movement stops when the up contact in the front window control switch is released. If the front window motor is removed from the window regulator drum housing, or if a new front window motor is installed, it must be initialized. Refer to Window Motor Initialization in this section.

The front window motors automatically adjust to system changes throughout its life: changes in seal drag and slight changes in the full OPEN position occur and are automatically compensated for. Once initialized, the window motor soft stalls into the upper position contributing to extended durability of the system. If the window does not seal completely in the full UP position (very small gaps/non bounce-back events only), the window switch can be actuated to the UP position to energize the window motor for a fraction of a second to fully seal and this new position is learned.

Bounce-Back

The front window motors have a bounce-back feature. If an obstacle has been detected in the window opening as the window glass is moving upward, the window motor automatically reverses direction and moves the glass toward the bounce-back position (in both up and one-touch up modes). If an obstruction occurs between 4 mm (0.15 in) and 200 mm (7.87 in) of window opening, the bounce-back position is approximately at 250 mm (9.84 in) of window opening. If an obstruction occurs at a position greater than 200 mm (7.87 in) of window opening, the bounce-back position is approximately at 50 mm (1.96 in) below where the obstruction occurred.

Security Override

To override a bounce-back condition (to overcome the resistance of ice on the window or seals for example), within 2 seconds after the window reaches the bounce-back position, pull and hold the window control switch. The window travels up with no bounce-back or pinch protection. If the window control switch is released before the window is fully closed, the window stops. If the ignition is turned to OFF or START (without delayed accessory), the window motor stops.

Rear Passenger Window Operation — SuperCab and SuperCrew Only

Rear passenger windows can be raised or lowered using the LH front window control switch or the corresponding passenger window control switch. The rear passenger window control switches receive power when the delayed accessory relay is active and the LH front window lock-out switch is in the UNLOCK position. When the LH front window lock-out switch is in the LOCK position, the rear passenger windows only operate from the LH front window control switch.

Rear Window Defrost (if equipped)

When the rear window defrost switch is pressed and the engine is running, the HVAC module activates the rear window defrost relay located in the Battery Junction Box (BJB). When the rear window defrost relay is active, the rear window defrost grid is energized and the rear window defrost status LED is illuminated. For vehicles equipped with a power sliding rear window, the rear window defrost grid does not activate unless the rear sliding window glass is in the fully CLOSED

position due to a glass position switch in series with the grid circuit. The glass position switch is integral to the power sliding rear window assembly. There is a feedback circuit incorporated into the glass position switch which illuminates the rear window defrost switch LED when the defrost grid is active and the sliding glass is fully closed. If the sliding glass is opened while the defrost system is active, the heated mirrors continue to operate and the rear window defrost switch LED is no longer illuminated.

The HVAC module deactivates the rear window defrost relay when one of the following conditions is met:

- The rear window defrost switch is pressed when the feature is active
- Ignition state is changed from RUN to OFF/LOCK
- The 15-minute timing function is completed

Feature Inputs

- Ignition RUN position (12V on both RUN and RUN/ACC inputs)
- Rear window defrost switch (momentary contact to ground when switch is pressed)

Feature Output

- Rear window defrost relay control (grounded when activated, open circuit when deactivated)

Remote Start Climate Operation (if equipped)

The customer can select different climate control modes/preferences when the vehicle is started using the remote start feature. This can be accessed through the message center. For information on how to set the remote start preferences, refer to the Owner's Literature. When the rear defrost is set to AUTO mode, the rear window defrost activates when the outside temperature is less than 0° C (32° F) and the vehicle is started using the remote start feature. No climate control adjustments are recognized during remote start operation. Once the ignition is cycled to the ON position, the climate control system returns to the previous settings (last ignition-ON cycle) and adjustments can be made normally. If the previous setting was off, the climate control system turns off.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection Chart

Mechanical	Electrical
• Window regulator and motor assembly • Window glass • Window control switch	• Loose, corroded connectors • Circuitry • Window control switch • Battery Junction Box (BJB) fuse(s): ▪ 32 (40A) ▪ 54 (5A) • Body Control Module (BCM) circuit breaker 48 (30A) • BCM fuse(s): ▪ 1 (30A) ▪ 3 (30A) ▪ 32 (15A) • Accessory delay relay • Rear window defrost relay • Rear window defrost grid

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:

- verify the ignition is in the ON position.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

7. Carry out the network test:

- If the scan tool responds with no communication from one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record the Continuous Memory Diagnostic Trouble Codes (CMDTCs) .

8. Clear the CMDTCs and carry out the self-test diagnostics for the BCM , FCIM and HVAC module.

9. If the DTCs retrieved are related to the concern, go to the Body Control Module (BCM) DTC Chart or HVAC Module DTC Chart. For all other DTCs, refer to Section 419-10.

10. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

Body Control Module (BCM) DTC Chart

DTC	Description	Action
B131A:12	Delayed Accessory Power: Circuit Short To Battery	GO to Pinpoint Test J.
B131A:14	Delayed Accessory Power: Circuit Short To Ground or Open	If the delayed accessory is always on, GO to Pinpoint Test K. If the delayed accessory is inoperative, GO to Pinpoint Test J.
All other DTCs	—	REFER to the Master DTC Chart in Section 419-10.

HVAC Module DTC Chart

DTC	Description	Action
B1C83:11	Rear Defog Relay: Circuit Short to Ground	REFER to Section 501-09 to diagnose the inoperative heated mirror system.
B1C83:12	Rear Defog Relay: Circuit Short to Battery	REFER to Section 501-09 to diagnose the inoperative heated mirror system.
B1C83:13	Rear Defog Relay: Circuit Open	REFER to Section 501-09 to diagnose the inoperative heated mirror system.
All other DTCs	—	REFER to the Master DTC Chart in Section 412-00.

Symptom Chart

Symptom Chart

Condition	Possible Causes	Action
All power windows are inoperative — intermittent operation, SuperCab or SuperCrew	- Circuit breaker - Wiring, terminals or connectors	GO to Pinpoint Test A.
All power windows are inoperative	- Fuse - Circuit breaker - Accessory delay relay - Wiring, terminals or connectors - Window control switch - Body Control Module (BCM)	GO to Pinpoint Test B.
All passenger power windows are inoperative — LH and RH rear	- Wiring, terminals or connectors - Window control switch	GO to Pinpoint Test L.
A single power window is inoperative — LH front	- Fuse - Wiring, terminals or connectors - Window control switch - Power window motor	GO to Pinpoint Test C.
A single power window is inoperative	Fuse	GO to Pinpoint Test D.

— RH front	• Wiring, terminals or connectors • Window control switch • Power window motor	
• A single power window is inoperative — LH rear	• Wiring, terminals or connectors • Window control switch • Power window motor	• GO to Pinpoint Test E.
• A single power window is inoperative — RH rear	• Wiring, terminals or connectors • Window control switch • Power window motor	• GO to Pinpoint Test F.
• A single power window is inoperative — power sliding rear window	• Wiring, terminals or connectors • Overhead console switch • Power sliding rear window motor	• GO to Pinpoint Test G.
• The one-touch up/down feature is inoperative	• Power window motor not initialized • Battery power was lost while power window motor was operating • Window control switch • Power window motor	• GO to Pinpoint Test M.
• The defrost system is inoperative	• Fuse(s) • Rear window defrost relay • Wiring, terminals or connectors • Rear sliding glass position switch (if equipped)	• GO to Pinpoint Test H.

	• Rear window defrost grid • HVAC module • Front Controls Interface Module (FCIM)	
• The defrost switch LED does not illuminate — the defroster operates	• Wiring, terminals or connectors • HVAC module • Front Controls Interface Module (FCIM) • Rear sliding glass contacts	• If equipped with rear sliding glass, GO to Pinpoint Test H. • If not equipped with rear sliding glass: • For vehicles equipped with EMTC , INSTALL a new HVAC module. REFER to Section 412-01. • For vehicles equipped with EATC , INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure.
• The defrost system does not shut off automatically	• Wiring, terminals or connectors • Rear window defrost relay • HVAC module	• GO to Pinpoint Test I.
• The defrost system does not respond to voice command only	• Voice recognition system	• VERIFY the rear window defrost operation using the instrument panel rear window defrost switch. If the rear window defrost is still inoperative, GO to Pinpoint Test H. If the rear window defrost operates, Refer to the appropriate section in Group 415 for the procedure.
• The defrost system does not respond to touchscreen command only	• Touchscreen	• VERIFY the rear window defrost operation using the instrument panel rear window defrost switch. If the rear window defrost is still inoperative, GO to Pinpoint Test H. If the rear window defrost operates, Refer to the appropriate section in Group 415 for the procedure.
• The delayed accessory is inoperative	• Door ajar switches • Wiring, terminals or connectors • Accessory delay relay • BCM	• GO to Pinpoint Test J.

• The delayed accessory does not turn off	• Wiring, terminals or connectors • Accessory delay relay • BCM	• GO to Pinpoint Test K.
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Pinpoint Tests

Pinpoint Test A: All Power Windows are Inoperative — Intermittent Operation, SuperCab or SuperCrew

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

Voltage is supplied to the LH front window control switch through Body Control Module (BCM) circuit breaker 48 (30A). Intermittent operation can be caused by a short to ground which temporarily causes the circuit breaker to open. Once open, all power windows are inoperative until the circuit breaker cools and the contacts close again. Activating all window control switches one at a time in both directions until the breaker opens can help in narrowing down the exact location of the short circuit if present.

This pinpoint test is intended to diagnose the following:

- Circuit breaker
- Wiring, terminals or connectors

PINPOINT TEST A : ALL POWER WINDOWS ARE INOPERATIVE — INTERMITTENT OPERATION

NOTICE: Use the correct probe adapter(s) when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

A1 CHECK FOR A SHORT TO GROUND IN THE POWER WINDOW SYSTEM

- Disconnect: ~~BCM~~ Circuit Breaker 48 (30A) .
- Install a fused jumper wire in place of ~~BCM~~ circuit breaker 48 (30A).
- Ignition ON.
- Attempt to operate each individual power window in both directions from the LH window control switch as well as the passenger door(s) window control switch(es).

Did the jumper wire fuse open during the test?

Yes	REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. TEST the system for normal operation.
No	INSTALL a new circuit breaker. TEST the system for normal operation.

Pinpoint Test B: All Power Windows are Inoperative

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

Voltage is supplied to the LH front window control switch from the accessory delay relay. For SuperCab or SuperCrew, the LH front window control switch supplies voltage to the rear window control switches when the lock-out switch is in the UNLOCK position.

This pinpoint test is intended to diagnose the following:

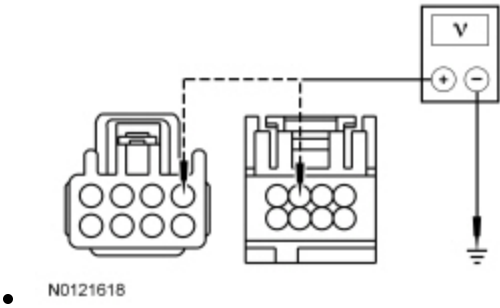
- Fuse
- Circuit breaker
- Accessory delay relay
- Wiring, terminals or connectors
- Window control switch
- BCM

PINPOINT TEST B : ALL POWER WINDOWS ARE INOPERATIVE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

B1 CHECK THE VOLTAGE TO THE LH FRONT WINDOW CONTROL SWITCH

- Disconnect: LH Front Window Control Switch [C504](#) (Regular Cab) or [C535A](#) (SuperCab or SuperCrew) .
- Ignition ON.
- Measure the voltage between LH front window control switch and ground.
 - For regular cab, measure the voltage between [C504](#) Pin 3, circuit CBP32 (GN/VT), harness side and ground.
 - For SuperCab or SuperCrew, measure the voltage between [C535B](#) Pin 1, circuit CBP01 (BU), harness side and ground.



Is the voltage greater than 10 volts?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	VERIFY Body Control Module (BCM) fuse 32 (15A) (regular cab) or BCM circuit breaker 48 (30A) (SuperCab or SuperCrew) is OK. If OK, GO to Pinpoint Test J to diagnose the accessory delay circuit. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test C: A Single Power Window is Inoperative — LH Front

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

NOTE: De-initialize, then initialize the front window motor when any of the following occur:

- the LH or RH front window motor is removed from the window regulator drum housing
- a new window regulator and/or motor is installed
- a new window glass is installed
- a new top run is installed
- a new top run is installed
- any operation in which grease or lubricants are applied to the window regulator or glass run

Normal Operation

Voltage is supplied to the LH front window control switch and LH front power window motor from the accessory delay relay. The LH front power window motor has dedicated high current voltage and ground circuits. When the LH front window control switch is activated, a low current voltage signal is supplied to the LH front power window motor which causes the window to move upward or downward.

This pinpoint test is intended to diagnose the following:

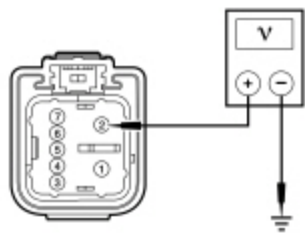
- Fuse
- Wiring, terminals or connectors
- Window control switch
- Power window motor

PINPOINT TEST C : A SINGLE POWER WINDOW IS INOPERATIVE — LH FRONT

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

C1 CHECK THE VOLTAGE TO THE POWER WINDOW MOTOR

- Ignition OFF.
- Disconnect: LH Front Power Window Motor [C524](#).
- Measure the voltage between LH front power window motor [C524](#) Pin 2, circuit SBP01 (RD), harness side and ground.



• N0121645

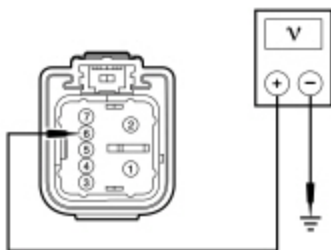
Is the voltage greater than 10 volts?

Yes	GO to C2 .
No	VERIFY Body Control Module (BCM) fuse 1 (30A) is OK. If OK, REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

C2 CHECK THE LH FRONT POWER WINDOW MOTOR DELAYED ACCESSORY FEED

- Ignition ON.

- Measure the voltage between LH front power window motor [C524](#) Pin 6, circuit CBP32 (GN/VT), harness side and ground.



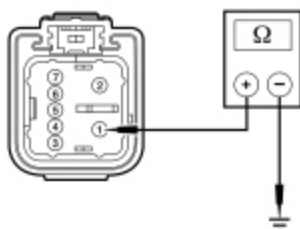
• N0121646

Is the voltage greater than 10 volts?

Yes	GO to C3 .
No	VERIFY Body Control Module (BCM) fuse 32 (15A) is OK. If OK, REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

C3 CHECK THE LH FRONT POWER WINDOW MOTOR GROUND

- Ignition OFF.
- Measure the resistance between LH front power window motor [C524](#) Pin 1, circuit GD134 (BK/VT), harness side and ground.



• N0121647

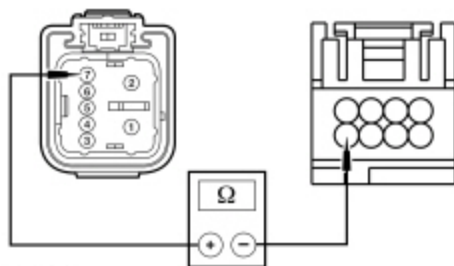
Is the resistance less than 5 ohms?

Yes	GO to C4 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

C4 CHECK THE WINDOW CONTROL SWITCH TO LH FRONT MOTOR CIRCUITS FOR AN OPEN

- Disconnect: LH Front Window Control Switch [C504](#) (regular cab) or [C535A](#) (SuperCab or SuperCrew) .
- Measure the resistance between LH front power window motor and LH front window control switch.

- For regular cab, measure the resistance between LH front power window motor [C524](#) Pin 7, circuit CPW10 (YE/VT), harness side and LH front window control switch [C504](#) Pin 8, circuit CPW10 (YE/VT), harness side.
- For SuperCab or SuperCrew, measure the resistance between LH front power window motor [C524](#) Pin 7, circuit CPW10 (YE/VT), harness side and LH front window control switch [C535A](#) Pin 8, circuit CPW10 (YE/VT), harness side.



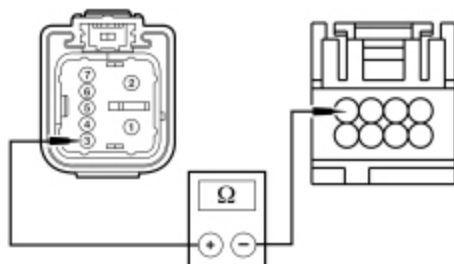
N0121649

Is the resistance less than 5 ohms?

Yes	GO to C5 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

C5 CHECK THE WINDOW CONTROL SWITCH TO LH FRONT MOTOR CIRCUITS FOR AN OPEN

- Measure the resistance between LH front power window motor and LH front window control switch.
 - For regular cab, measure the resistance between LH front power window motor [C524](#) Pin 3, circuit CPW11 (BU/GY), harness side and LH front window control switch [C504](#) Pin 4, circuit CPW11 (BU/GY), harness side.
 - For SuperCab or SuperCrew, measure the resistance between LH front power window motor [C524](#) Pin 3, circuit CPW11 (BU/GY), harness side and LH front window control switch [C535A](#) Pin 4, circuit CPW11 (BU/GY), harness side.



N0121650

Is the resistance less than 5 ohms?

Yes	GO to C6 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

C6 CHECK THE LH FRONT WINDOW CONTROL SWITCH

- Carry out the LH front window control switch component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the window control switch pass the component test?

Yes	INSTALL a new LH front window regulator motor. REFER to Window Regulator Motor in this section. INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.
No	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. INITIALIZE the LH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

Pinpoint Test D: A Single Power Window is Inoperative — RH Front

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

NOTE: *De-initialize, then initialize the front power window motor when any of the following occur:*

- *the LH or RH front power window motor is removed from the window regulator drum housing*
- *a new window regulator and/or motor is installed*
- *a new window glass is installed*
- *a new top run is installed*
- *a new top run is installed*
- *any operation in which grease or lubricants are applied to the window regulator or glass run*

Normal Operation

Voltage is supplied to the LH and RH front window control switches and RH front power window motor from the accessory delay relay. The RH front power window motor has dedicated high current voltage and ground circuits. When the LH or RH front window control switch is activated, a low current voltage signal is supplied to the RH front power window motor which causes the window to move upward or downward.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Window control switch
- Power window motor

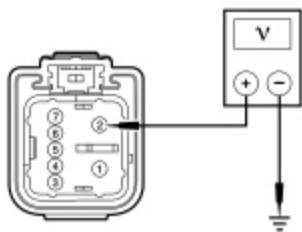
PINPOINT TEST D : A SINGLE POWER WINDOW IS INOPERATIVE — RH FRONT

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
D1 CHECK THE RH FRONT POWER WINDOW OPERATION FROM THE LH FRONT WINDOW CONTROL SWITCH
• Ignition ON. • Attempt to operate the RH front window using the LH front window control switch.
Does the RH front window operate when using the LH front window control switch?

Yes	GO to D8 .
No	GO to D2 .

D2 CHECK THE VOLTAGE TO THE POWER WINDOW MOTOR

- Ignition OFF.
- Disconnect: RH Front Power Window Motor [C608](#).
- Measure the voltage between RH front power window motor [C608](#) Pin 2, circuit SBP03 (BU/RD), harness side and ground.



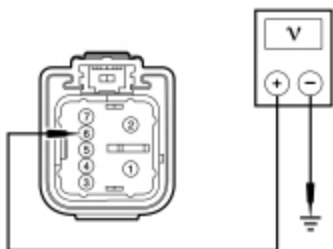
• N0121645

Is the voltage greater than 10 volts?

Yes	GO to D3 .
No	VERIFY Body Control Module (BCM) fuse 3 (30A) is OK. If OK, REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D3 CHECK THE RH FRONT POWER WINDOW MOTOR DELAYED ACCESSORY FEED

- Ignition ON.
- Measure the voltage between RH front power window motor [C608](#) Pin 6, circuit CBP32 (GN/VT), harness side and ground.



• N0121646

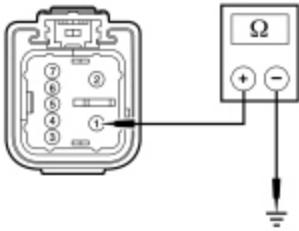
Is the voltage greater than 10 volts?

Yes	GO to D4 .
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No	VERIFY Body Control Module (BCM) fuse 32 (15A) is OK. If OK, REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.
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D4 CHECK THE RH FRONT POWER WINDOW MOTOR GROUND

- Ignition OFF.
- Measure the resistance between RH front power window motor [C608](#) Pin 1, circuit GD139 (BK/YE), harness side and ground.



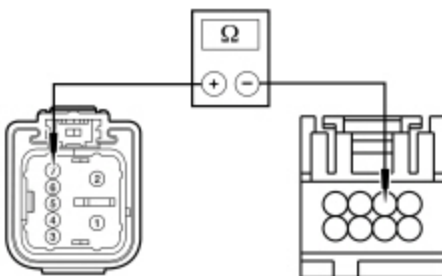
• N0121647

Is the resistance less than 5 ohms?

Yes	GO to D5 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D5 CHECK THE WINDOW CONTROL SWITCH TO RH FRONT MOTOR CIRCUITS FOR AN OPEN

- Disconnect: LH Front Window Control Switch [C504](#) (Regular Cab) or [C535A](#) (SuperCab or SuperCrew) .
- Measure the resistance between RH front power window motor and LH front window control switch.
 - For regular cab, measure the resistance between RH front power window motor [C608](#) Pin 7, circuit CPW12 (GN/OG), harness side and LH front window control switch [C504](#) Pin 2, circuit CPW12 (GN/OG), harness side.
 - For SuperCab or SuperCrew, measure the resistance between RH front power window motor [C608](#) Pin 7, circuit CPW12 (GN/OG), harness side and LH front window control switch [C535A](#) Pin 2, circuit CPW12 (GN/OG), harness side.



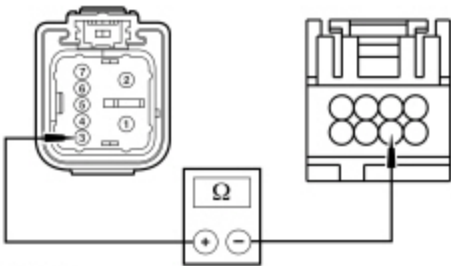
• N0121652

Is the resistance less than 5 ohms?

Yes	GO to D6 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D6 CHECK THE WINDOW CONTROL SWITCH TO RH FRONT MOTOR CIRCUITS FOR AN OPEN

- Measure the resistance between RH front power window motor and LH front window control switch.
 - For regular cab, measure the resistance between RH front power window motor [C608](#) Pin 3, circuit CPW13 (BN/YE), harness side and LH front window control switch [C504](#) Pin 6, circuit CPW13 (BN/YE), harness side.
 - For SuperCab or SuperCrew, measure the resistance between RH front power window motor [C608](#) Pin 3, circuit CPW13 (BN/YE), harness side and LH front window control switch [C535A](#) Pin 6, circuit CPW13 (BN/YE), harness side.



• N0121654

Is the resistance less than 5 ohms?

Yes	GO to D7 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D7 CHECK THE LH FRONT WINDOW CONTROL SWITCH

- Carry out the LH front window control switch component test. Refer to Wiring Diagrams Cell [149](#) for component testing.

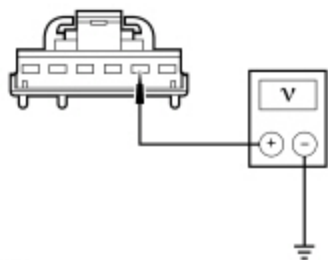
Did the window control switch pass the component test?

Yes	INSTALL a new RH front window regulator motor. REFER to Window Regulator Motor in this section. INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.
No	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D8 CHECK THE VOLTAGE TO THE RH FRONT WINDOW CONTROL SWITCH

- Ignition OFF.
- Disconnect: RH Front Window Control Switch [C604](#).

- Ignition ON.
- Measure the voltage between RH front window control switch [C604](#) Pin 2, circuit CBP32 (GN/VT), harness side and ground.



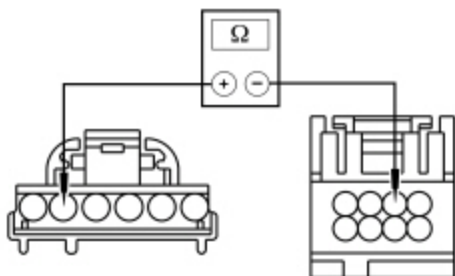
• N0043634

Is the voltage greater than 10 volts?

Yes	GO to D9 .
No	REPAIR the circuit. TEST the system for normal operation.

D9 CHECK THE WINDOW CONTROL SWITCH CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Front Window Control Switch [C504](#) (Regular Cab) or [C535A](#) (SuperCab or SuperCrew) .
- Measure the resistance between RH front window control switch and LH front window control switch.
 - For regular cab, measure the resistance between RH front window control switch [C604](#) Pin 5, circuit CPW12 (GN/OG), harness side and LH front window control switch [C504](#) Pin 2, circuit CPW12 (GN/OG), harness side.
 - For SuperCab or SuperCrew, measure the resistance between RH front window control switch [C604](#) Pin 5, circuit CPW12 (GN/OG), harness side and LH front window control switch [C535A](#) Pin 2, circuit CPW12 (GN/OG), harness side.



• N0121685

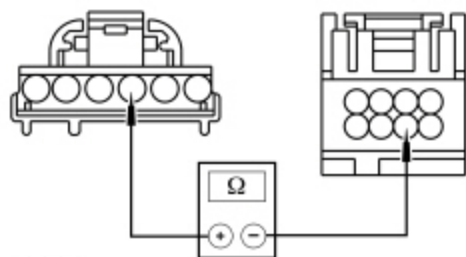
Is the resistance less than 5 ohms?

Yes	GO to D10 .
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

D10 CHECK THE WINDOW CONTROL SWITCH CIRCUITS FOR AN OPEN

- Measure the resistance between RH front window control switch and LH front window control switch.

- For regular cab, measure the resistance between RH front window control switch [C604](#) Pin 3, circuit CPW13 (BN/YE), harness side and LH front window control switch [C504](#) Pin 6, circuit CPW13 (BN/YE), harness side.
- For SuperCab or SuperCrew, measure the resistance between RH front window control switch [C604](#) Pin 3, circuit CPW13 (BN/YE), harness side and LH front window control switch [C535A](#) Pin 6, circuit CPW13 (BN/YE), harness side.



N0121685

Is the resistance less than 5 ohms?

Yes	INSTALL a new RH front window control switch. REFER to Window Control Switch — Front in this section. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.
No	REPAIR the circuit. DE-INITIALIZE, then INITIALIZE the RH front power window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.

Pinpoint Test E: A Single Power Window is Inoperative — LH Rear

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The LH rear window control switch receives voltage from the LH front window control switch when the lock-out switch is in the UNLOCK position. Ground is also provided to the LH rear window control switch from the LH front window control switch. The LH rear window control switch transfers voltage and ground to the power window motor. Window direction is controlled by reversing polarity of the voltage and ground circuits being supplied to the power window motor.

When operating the LH rear window from the LH front window control switch, voltage and ground is sent from the LH front window control switch through the normally closed contacts within the LH rear window control switch. Window direction is controlled by reversing polarity of the voltage and ground circuits being supplied to the power window motor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Window control switch
- Power window motor

PINPOINT TEST E : A SINGLE POWER WINDOW IS INOPERATIVE — LH REAR

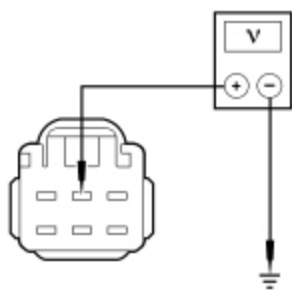
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

E1 CHECK THE VOLTAGE TO THE LH REAR WINDOW CONTROL SWITCH

- Disconnect: LH Rear Window Control Switch [C701](#).

- Ignition ON.
- **NOTE:** Set the LH front window lock-out switch to the *UNLOCK* position.

Measure the voltage between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and ground.



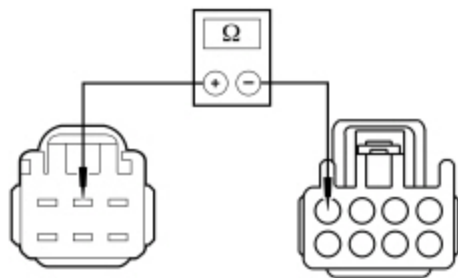
N0092722

Is the voltage greater than 10 volts?

Yes	GO to E3 .
No	GO to E2 .

E2 CHECK THE WINDOW CONTROL SWITCH POWER CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and LH front window control switch [C535B](#) Pin 4, circuit CPW14 (VT/WH), harness side.



N0121692

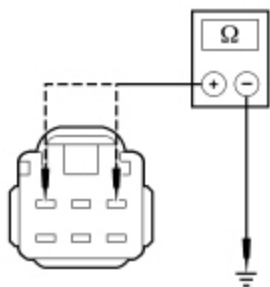
Is the resistance less than 5 ohms?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

E3 CHECK THE GROUNDS TO THE LH REAR WINDOW CONTROL SWITCH

- Ignition OFF.
- Measure the resistance between LH rear window control switch [C701](#) Pin 1, circuit CPW16 (BU/OG), harness side and ground; and between LH rear window control switch [C701](#) Pin 3, circuit CPW15 (YE), harness side

and ground.



N0092724

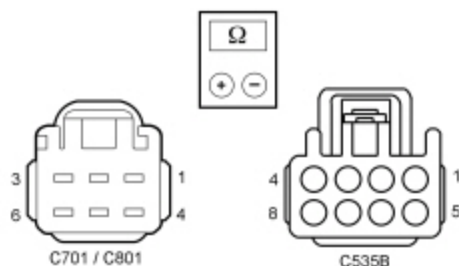
Are the resistances less than 5 ohms?

Yes	GO to E5 .
No	GO to E4 .

E4 CHECK THE CIRCUITS TO THE LH REAR WINDOW CONTROL SWITCH FOR AN OPEN

- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between LH rear window control switch [C701](#), harness side and LH front window control switch [C535B](#), harness side using the following chart:

LH Rear Window Control Switch	Circuit	LH Front Window Control Switch
C701 Pin 3	CPW15 (YE)	C535B Pin 6
C701 Pin 1	CPW16 (BU/OG)	C535B Pin 5



N0121693

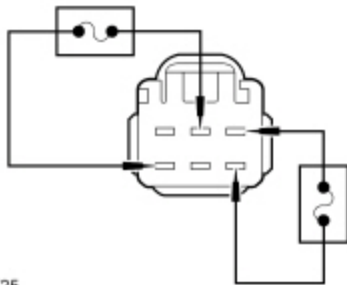
Are the resistances less than 5 ohms?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). TEST the system for normal operation.

E5 CHECK THE LH REAR WINDOW OPERATION

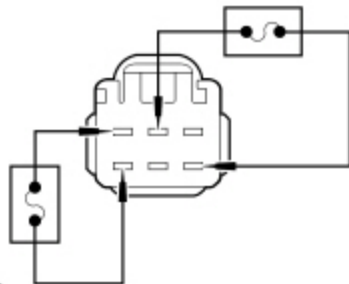
- Ignition ON. **NOTE:** Set the LH front window lock-out switch to the UNLOCK position.

- Connect a fused jumper wire between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and [C701](#) Pin 6, circuit CPW21 (BN), harness side; and a second fused jumper wire between LH rear window control switch [C701](#) Pin 4, circuit CPW22 (GN/VT), harness side and [C701](#) Pin 1, circuit CPW16 (BU/OG), harness side. The LH rear window should operate in the upward direction.



N0092725

- Connect a fused jumper wire between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and [C701](#) Pin 4, circuit CPW22 (GN/VT), harness side; and a second fused jumper wire between LH rear window control switch [C701](#) Pin 6, circuit CPW21 (BN), harness side and [C701](#) Pin 3, circuit CPW15 (YE), harness side. The LH rear window should operate in the downward direction.



N0092726

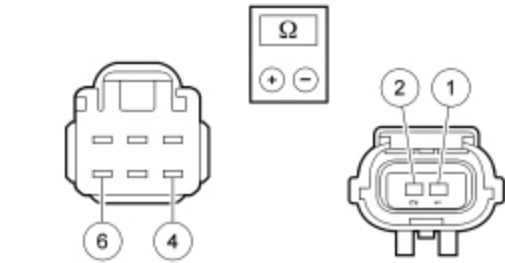
Does the LH rear window operate correctly?

Yes	If the LH rear window did not operate with the LH rear window control switch, INSTALL a new LH rear window control switch. REFER to Window Control Switch — Rear in this section. TEST the system for normal operation. If the LH rear window did operate with the LH rear window control switch, but not with the LH front window control switch, INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	GO to E6 .

E6 CHECK THE POWER WINDOW MOTOR CIRCUITS FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Rear Power Window Motor [C726](#) (SuperCab) or [C703](#) (SuperCrew) .
- For SuperCab, measure the resistance between LH rear window control switch [C701](#), harness side and LH rear power window motor [C726](#), harness side using the following chart:

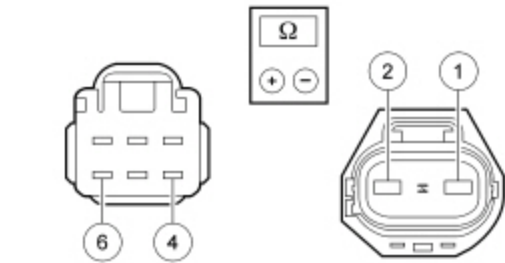
LH Rear Window Control Switch	Circuit	LH Rear Power Window Motor
C701 Pin 6	CPW21 (BN)	C726 Pin 1
C701 Pin 4	CPW22 (GN/VT)	C726 Pin 2



• N0092730

- For SuperCrew, measure the resistance between LH rear window control switch [C701](#), harness side and LH rear power window motor [C703](#), harness side using the following chart:

LH Rear Window Control Switch	Circuit	LH Rear Power Window Motor
C701 Pin 6	CPW21 (BN)	C703 Pin 1
C701 Pin 4	CPW22 (GN/VT)	C703 Pin 2



• N0092731

Are the resistances less than 5 ohms?

Yes	INSTALL a new LH rear power window motor. REFER to Window Regulator Motor in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). TEST the system for normal operation.

Pinpoint Test F: A Single Power Window is Inoperative — RH Rear

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The RH rear window control switch receives voltage from the LH front window control switch when the lock-out switch is in the UNLOCK position. Ground is also provided to the RH rear window control switch from the LH front window control switch. The RH rear window control switch transfers voltage and ground to the power window motor. Window direction is controlled by reversing polarity of the voltage and ground circuits being supplied to the power window motor.

When operating the RH rear window from the LH front window control switch, voltage and ground is sent from the LH front window control switch through the normally closed contacts within the RH rear window control switch. Window direction is controlled by reversing polarity of the voltage and ground circuits being supplied to the power window motor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Window control switch
- Power window motor

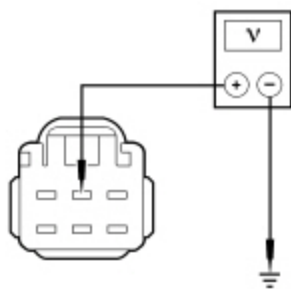
PINPOINT TEST F : A SINGLE POWER WINDOW IS INOPERATIVE — RH REAR

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

F1 CHECK THE VOLTAGE TO THE RH REAR WINDOW CONTROL SWITCH

- Disconnect: RH Rear Window Control Switch [C801](#).
- Ignition ON.
- **NOTE:** Set the LH front window lock-out switch to the UNLOCK position.

Measure the voltage between RH rear window control switch [C801](#) Pin 2, circuit CPW14 (VT/WH), harness side and ground.

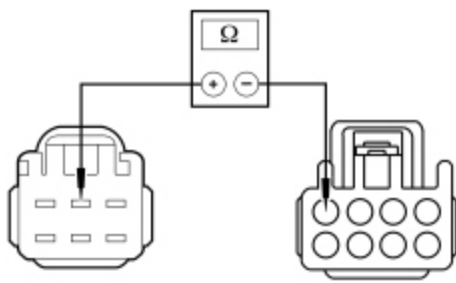


Is the voltage greater than 10 volts?

Yes	GO to F3 .
No	GO to F2 .

F2 CHECK THE WINDOW CONTROL SWITCH POWER CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between RH rear window control switch [C801](#) Pin 2, circuit CPW14 (VT/WH), harness side and LH front window control switch [C535B](#) Pin 4, circuit CPW14 (VT/WH), harness side.



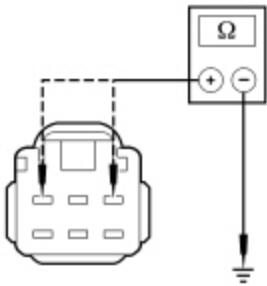
• N0121692

Is the resistance less than 5 ohms?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

F3 CHECK THE GROUNDS TO THE LH REAR WINDOW CONTROL SWITCH

- Ignition OFF.
- Measure the resistance between RH rear window control switch [C801](#) Pin 1, circuit CPW18 (GY/VT), harness side and ground; and between RH rear window control switch [C801](#) Pin 3, circuit CPW17 (BN/GN), harness side and ground.



• N0092724

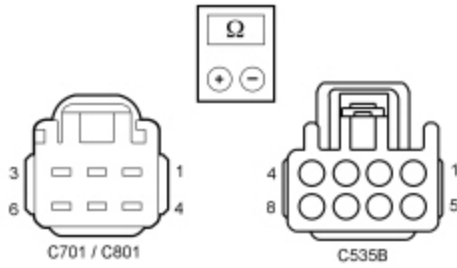
Are the resistances less than 5 ohms?

Yes	GO to F5 .
No	GO to F4 .

F4 CHECK CIRCUITS CPW17 (BN/GN) AND CPW18 (GY/VT) FOR AN OPEN

- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between RH rear window control switch [C801](#), harness side and LH front window control switch [C535B](#), harness side using the following chart:

RH Rear Window Control Switch	Circuit	LH Front Window Control Switch
C801 Pin 3	CPW17 (BN/GN)	C535B Pin 4
C801 Pin 1	CPW18 (GY/VT)	C535B Pin 7



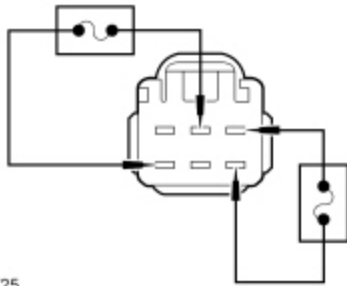
N0121693

Are the resistances less than 5 ohms?

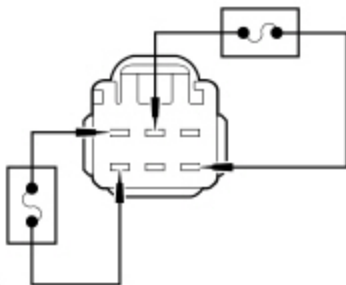
Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). TEST the system for normal operation.

F5 CHECK THE RH REAR WINDOW OPERATION

- Ignition ON. **NOTE:** Set the LH front window lock-out switch to the UNLOCK position.
- Connect a fused jumper wire between RH rear window control switch [C801](#) Pin 2, circuit CPW14 (VT/WH), harness side and [C801](#) Pin 6, circuit CPW23 (GY), harness side; and a second fused jumper wire between RH rear window control switch [C801](#) Pin 4, circuit CPW24 (WH/VT), harness side and [C801](#) Pin 1, circuit CPW18 (GY/VT), harness side. The RH rear window should operate in the downward direction.



- Connect a fused jumper wire between RH rear window control switch [C801](#) Pin 2, circuit CPW14 (VT/WH), harness side and [C801](#) Pin 4, circuit CPW24 (WH/VT), harness side; and a second fused jumper wire between RH rear window control switch [C801](#) Pin 6, circuit CPW23 (GY), harness side and [C801](#) Pin 3, circuit CPW17 (BN/GN), harness side. The RH rear window should operate in the upward direction.



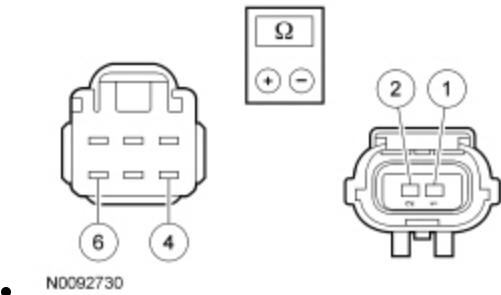
Does the LH rear window operate correctly?

Yes	If the RH rear window did not operate with the RH rear window control switch, INSTALL a new RH rear window control switch. REFER to Window Control Switch — Rear in this section. TEST the system for normal operation. If the RH rear window did operate with the RH rear window control switch, but not with the LH front window control switch, INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	GO to F6 .

F6 CHECK THE POWER WINDOW MOTOR CIRCUITS FOR AN OPEN

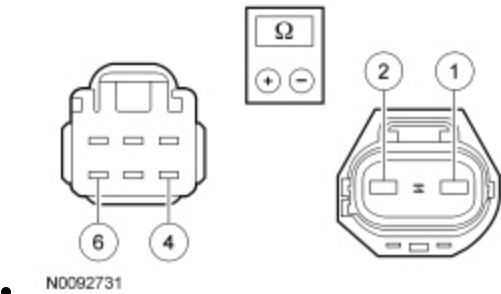
- Ignition OFF.
- Disconnect: RH Rear Power Window Motor [C828](#) (SuperCab) or [C803](#) (SuperCrew) .
- For SuperCab, measure the resistance between RH rear window control switch [C801](#), harness side and RH rear power window motor [C828](#), harness side using the following chart:

RH Rear Window Control Switch	Circuit	RH Rear Power Window Motor
C801 Pin 6	CPW23 (GY)	C828 Pin 1
C801 Pin 4	CPW24 (WH/VT)	C828 Pin 2



- For SuperCrew, measure the resistance between RH rear window control switch [C801](#), harness side and RH rear power window motor [C803](#), harness side using the following chart:

RH Rear Window Control Switch	Circuit	RH Rear Power Window Motor
C801 Pin 6	CPW23 (GY)	C803 Pin 1
C801 Pin 4	CPW24 (WH/VT)	C803 Pin 2



Are the resistances less than 5 ohms?

Yes	INSTALL a new RH rear power window motor. REFER to Window Regulator Motor in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). TEST the system for normal operation.

Pinpoint Test G: A Single Power Window is Inoperative — Power Sliding Rear Window

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The overhead console switch supplies voltage and ground to the power sliding window motor to control movement. The overhead console switch controls movement by reversing polarity of the voltage and ground circuits being supplied to the power sliding window motor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Overhead console switch
- Power sliding window motor

PINPOINT TEST G : A SINGLE POWER WINDOW IS INOPERATIVE — POWER SLIDING REAR WINDOW

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

G1 CHECK THE POWER WINDOW OPERATION

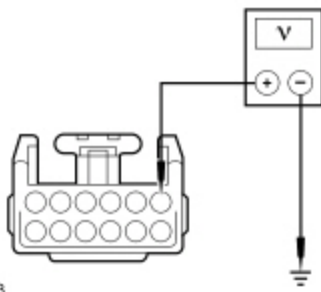
- Ignition ON.
- Verify the power windows operate.

Do the power windows operate?

Yes	GO to G2 .
No	GO to Pinpoint Test B to diagnose the inoperative power windows.

G2 CHECK THE VOLTAGE TO THE OVERHEAD CONSOLE SWITCH

- Ignition OFF.
- Disconnect: Overhead Console Switch [C912](#).
- Ignition ON.
- Measure the voltage between overhead console switch [C912](#) Pin 1, circuit CBP01 (BU), harness side and ground.



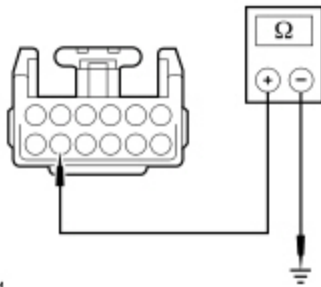
• N0092733

Is the voltage greater than 10 volts?

Yes	GO to G3 .
No	REPAIR the circuit. TEST the system for normal operation.

G3 CHECK THE GROUND TO THE OVERHEAD CONSOLE SWITCH FOR AN OPEN

- Ignition OFF.
- Measure the resistance between overhead console switch [C912](#) Pin 11, circuit GD139 (BK/YE), harness side and ground.



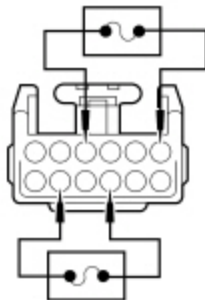
• N0092734

Is the resistance less than 5 ohms?

Yes	GO to G4 .
No	REPAIR the circuit(s). TEST the system for normal operation.

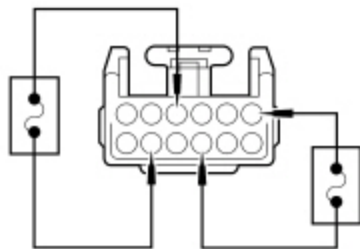
G4 CHECK THE OVERHEAD CONSOLE SWITCH

- Ignition ON.
- Connect a fused jumper between overhead console switch [C912](#) Pin 1, circuit CBP01 (BU), harness side and [C912](#) Pin 4, circuit CPW59 (VT/BN), harness side; and a second fused jumper wire between overhead console switch [C912](#) Pin 9, circuit CPW58 (YE/OG), harness side and [C912](#) Pin 11, circuit GD139 (BK/YE), harness side. The power sliding window should open.



• N0092735

- Connect a fused jumper between overhead console switch [C912](#) Pin 1, circuit CBP01 (BU), harness side and [C912](#) Pin 9, circuit CPW58 (YE/OG), harness side; and a second fused jumper wire between overhead console switch [C912](#) Pin 4, circuit CPW59 (VT/BN) and [C912](#) Pin 11, circuit GD139 (BK/YE), harness side. The rear sliding window should close.



• N0092736

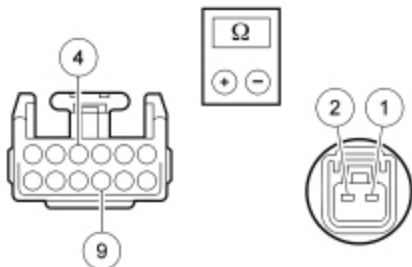
Did the rear power sliding window open and close?

Yes	INSTALL a new overhead console control switch. TEST the system for normal operation.
No	GO to G5 .

G5 CHECK THE POWER SLIDING WINDOW MOTOR CIRCUITS FOR AN OPEN

- Disconnect: Power Sliding Window Motor [C3145](#).
- Measure the resistance between overhead console switch [C912](#), harness side and power sliding window motor [C3145](#), harness side using the following chart:

Overhead Console Switch	Circuit	Power Sliding Window Motor
C912 Pin 9	CPW58 (YE/OG)	C3145 Pin 2
C912 Pin 4	CPW59 (VT/BN)	C3145 Pin 1



• N0092737

Are the resistances less than 5 ohms?

Yes	INSTALL a new rear sliding window motor. REFER to Window Regulator Motor — Rear in this section. TEST the system for normal operation.
No	REPAIR the circuit(s). TEST the system for normal operation.

Pinpoint Test H: The Defrost System is Inoperative

Refer to Wiring Diagrams Cell [56](#) , Heated Window for schematic and connector information.

Normal Operation

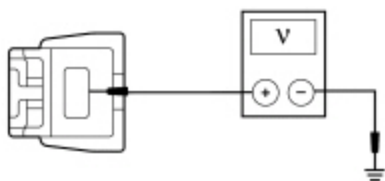
When the engine is running and the defrost switch is pressed, the HVAC module activates the rear window defrost relay. The defrost system is activated at one of the three following locations: HVAC module, FCIM or FDIM (8-inch [203mm] touchscreen). The FCIM and/or FDIM sends the defrost request message to the HVAC module over the Controller Area Network (CAN) . When the rear window defrost relay is active, voltage is supplied to the rear window defrost grid. For vehicles equipped with a power sliding rear window, the rear window defrost grid does not activate unless the rear sliding window glass is in the fully CLOSED position due to a glass position switch in series with the grid circuit. The glass position switch is integral to the power sliding rear window assembly. The glass position switch sends a feedback signal to the HVAC module. When this signal is received, the HVAC module sends a message to the FCIM to illuminate the rear window defrost switch LED when the defrost grid is active and the sliding glass is fully closed. If the sliding glass is opened while the defrost system is active, the heated mirrors continue to operate and the rear window defrost switch LED is no longer illuminated.

This pinpoint test is intended to diagnose the following:

- Fuse(s)
- Rear window defrost relay
- Wiring, terminals or connectors
- Rear sliding glass position switch (if equipped) (part of power sliding rear window assembly)
- Rear window defrost grid
- HVAC module
- FCIM

PINPOINT TEST H : THE DEFROST SYSTEM IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
H1 CHECK THE REAR WINDOW DEFROST GRID VOLTAGE
• Disconnect: Rear Window Defrost Grid Power (passenger side outer terminal) . • Start the engine. • Activate the rear window defrost switch and measure the voltage between rear window defrost grid power terminal, harness side and ground.



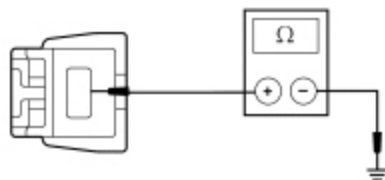
• A0010641

Is the voltage greater than 10 volts?

Yes	GO to H2 .
No	GO to H5 .

H2 CHECK THE REAR WINDOW DEFROST GRID GROUND

- Ignition OFF.
- Disconnect: Rear Window Defrost Grid Ground (Driver Side Terminal) .
- Measure the resistance between rear window defrost grid ground terminal, harness side and ground.



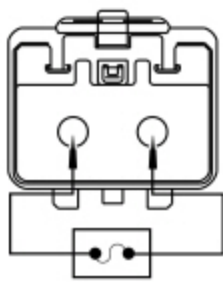
• A0010644

Is the resistance less than 5 ohms?

Yes	If equipped with a power sliding rear window, GO to H3 . If not equipped with a power sliding rear window, CARRY OUT the Grid Wire Test in this section. ATTEMPT to REPAIR the rear window defrost grid, REFER to Window Grid Wire Repair. If not repairable, INSTALL a new rear window glass assembly. REFER to Window Glass — Rear in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

H3 CHECK THE REAR WINDOW DEFROST SWITCH LED OPERATION

- Ignition OFF.
- Disconnect: Rear Window Defrost Grid Inline (Passenger Side) [C3368](#).
- Install a fused jumper wire between rear window defrost grid inline [C3368](#) Pin 1, circuit CH122 (WH/OG), harness side and [C3368](#) Pin 2, circuit CRD06 (BN/YE), harness side.



• N0122405

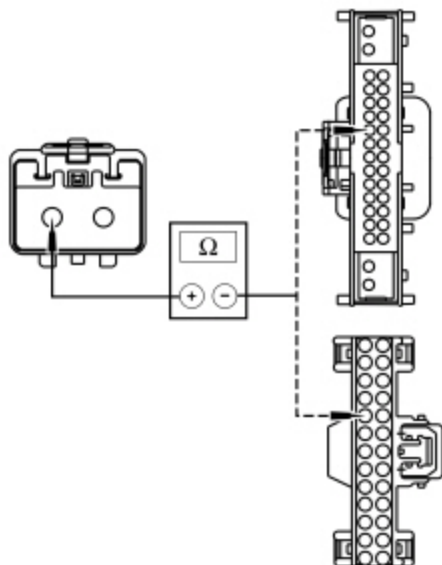
- Start the engine.
- Activate the rear window defrost switch.

Does the rear window defrost switch LED illuminate?

Yes	CARRY OUT the Grid Wire Test in this section. ATTEMPT to REPAIR the rear window defrost grid, REFER to Window Grid Wire Repair. If not repairable, INSTALL a new rear window glass assembly. REFER to Window Glass — Rear in this section. TEST the system for normal operation.
No	GO to H4 .

H4 CHECK THE REAR WINDOW DEFROST FEEDBACK CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: HVAC Module — EATC [C228A](#) or HVAC Module — Electronic Manual Temperature Control (EMTC) [C294A](#).
- Measure the resistance between rear window defrost grid inline [C3368](#) and the HVAC module.
 - For vehicles equipped with EATC, measure the resistance between rear window defrost grid inline [C3368](#) Pin 1, circuit CH122 (WH/OG), harness side and HVAC module — EATC [C294A](#) Pin 22, circuit CH122 (WH/OG), harness side.
 - For vehicles equipped with EATC, measure the resistance between rear window defrost grid inline [C3368](#) Pin 1, circuit CH122 (WH/OG), harness side and HVAC module — EATC [C228A](#) Pin 7, circuit CH122 (WH/OG), harness side.



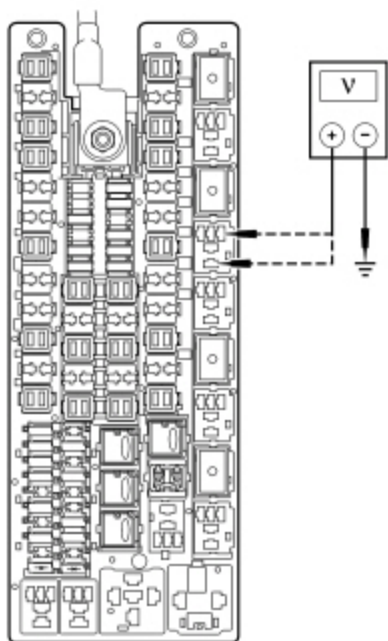
• N0148574

Is the resistance less than 5 ohms?

Yes	GO to H10 .
No	REPAIR the circuit. TEST the system for normal operation.

H5 CHECK THE POWER TO THE RELAY

- Ignition OFF.
- Disconnect: Rear Window Defrost Relay .
- Ignition ON.
- Measure the voltage between rear window defrost relay socket pin 1, circuit CBB54 (VT/OG), harness side and ground; and between pin 3, circuit SBB32 (VT/RD), harness side and ground.



• N0093251

Is the voltage greater than 10 volts for both measurements?

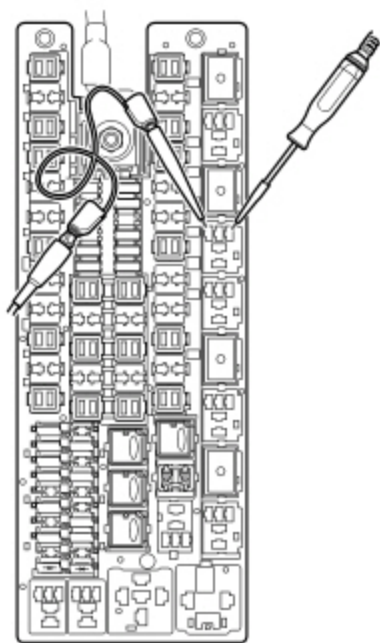
Yes	GO to H6 .
No	VERIFY Battery Junction Box (BJB) fuse(s) 32 (40A) and 54 (5A) are OK. If OK, REPAIR the circuit(s). TEST the system for normal operation. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. TEST the system for normal operation.

H6 CHECK THE HVAC MODULE

NOTICE: The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.

- Start the engine.

- Activate the rear window defrost switch.
- Connect the 12-volt incandescent test light between rear window defrost relay socket pin 2, circuit CRD02 (BN/BU) and rear window defrost relay socket pin 1, circuit CBB54 (VT/OG).



• N0101517

Does the test light illuminate?

Yes	GO to H7 .
No	For vehicles equipped with the 4.2-inch (107mm) or 8-inch (203mm) display, GO to H8 . For vehicles not equipped with the 4.2-inch (107mm) or 8-inch (203mm) display, GO to H9 .

H7 CHECK THE REAR WINDOW DEFROST RELAY

- Ignition OFF.
- Carry out the rear window defrost relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the rear window defrost relay pass the component test?

Yes	REPAIR open in circuit CRD06 (BN/YE). TEST the system for normal operation.
No	INSTALL a new rear window defrost relay. TEST the system for normal operation.

H8 CHECK THE REAR DEFROST SWITCH (CC_SW_R_DEF) PID

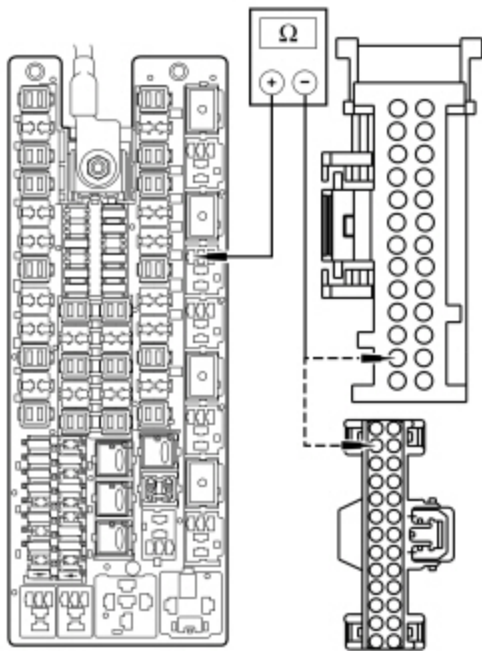
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — **FCIM** .
- Monitor the **FCIM** CC_SW_R_DEF PID while pressing and releasing the rear window defrost switch.

Does the PID agree with the rear defrost switch status?

Yes	GO to H9 .
No	INSTALL a new FCIM . Refer to the appropriate section in Group 415 for the procedure. TEST the system for normal operation.

H9 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: HVAC Module — EATC [C228A](#) or HVAC Module — Electronic Manual Temperature Control (EMTC) [C294A](#).
- Measure the resistance between rear window defrost relay and the HVAC module.
 - For vehicles equipped with EATC , measure the resistance between rear window defrost relay socket pin 2, circuit CRD02 (BN/BU) and HVAC module — EMTC [C294A](#) Pin 25, circuit CRD02 (BN/BU), harness side.
 - For vehicles equipped with EATC , measure the resistance between rear window defrost relay socket pin 2, circuit CRD02 (BN/BU) and HVAC module — EATC [C228A](#) Pin 25, circuit CRD02 (BN/BU), harness side.



N0148575

Is the resistance less than 5 ohms?

Yes	GO to H10 .
No	REPAIR the circuit. TEST the system for normal operation.

H10 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect all HVAC module electrical connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all HVAC module electrical connectors and make sure they seat correctly.

- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test I: The Defrost System Does Not Shut Off Automatically

Refer to Wiring Diagrams Cell [56](#) , Heated Window for schematic and connector information.

Normal Operation

When the engine is running and the defrost switch is pressed, the HVAC module activates the rear window defrost relay. The rear window defrost switch is integral to the HVAC module (if equipped with E.M.T.C.) or the F.C.I.M (if equipped with E.A.T.C). For vehicles equipped with E.A.T.C , the F.C.I.M sends the defrost request message to the HVAC module over the Controller Area Network (CAN) . When the rear window defrost relay is active, voltage is supplied to the rear window defrost grid. For vehicles equipped with a power sliding rear window, the rear window defrost grid does not activate unless the rear sliding window glass is in the fully CLOSED position due to a glass position switch in series with the grid circuit. The glass position switch is integral to the power sliding rear window assembly. The glass position switch sends a feedback signal to the HVAC module to illuminate the rear window defrost switch LED when the defrost grid is active and the sliding glass is fully closed. For vehicles equipped with E.A.T.C , the HVAC module sends a message to the F.C.I.M to illuminate the rear window defrost switch LED when the defrost grid is active and the sliding glass is fully closed. If the sliding glass is opened while the defrost system is active, the heated mirrors continue to operate and the rear window defrost switch LED is no longer illuminated.

This pinpoint test is intended to diagnose the following:

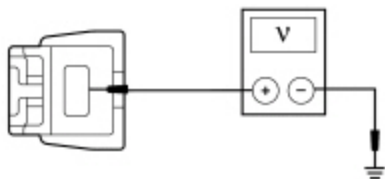
- Wiring, terminals or connectors
- Rear window defrost relay
- HVAC module

PINPOINT TEST I : THE DEFROST SYSTEM WILL NOT SHUT OFF AUTOMATICALLY

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

I1 CHECK REAR WINDOW DEFROST GRID POWER CIRCUIT FOR A SHORT TO VOLTAGE

- Disconnect: Rear Window Defrost Relay .
- Disconnect: Rear Window Defrost Grid Power (passenger side outer terminal) .
- Ignition ON.
- Measure the voltage between rear window defrost grid power terminal, harness side and ground.



• A0010641

Is the voltage greater than 10 volts?

Yes	REPAIR circuit CRD06 (BN/YE) for a short to voltage. TEST the system for normal operation.
No	If equipped with a power sliding rear window, GO to I2 . If not equipped with a power sliding rear window, GO to I4 .

I2 CHECK THE REAR WINDOW DEFROST SWITCH LED STATUS

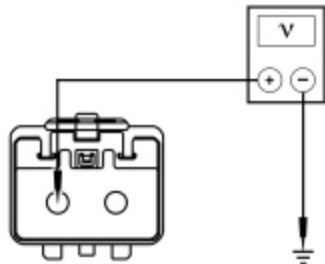
- Observe the rear window defrost switch status.

Is the rear window defrost switch LED illuminated?

Yes	GO to I3 .
No	GO to I4 .

I3 CHECK THE REAR WINDOW DEFROST FEEDBACK CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: Rear Window Defrost Grid Inline (passenger side) [C3368](#).
- Disconnect: HVAC Module — EATC [C228A](#) or HVAC Module — Electronic Manual Temperature Control (EMTC) [C294A](#).
- Ignition ON.
- Measure the voltage between rear window defrost grid inline [C3368](#) Pin 1, circuit CH122 (WH/OG), harness side and ground.



• N0122474

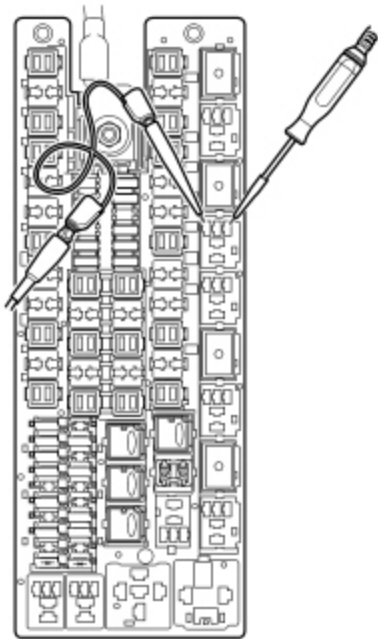
Is any voltage present?

Yes	REPAIR the circuit. TEST the system for normal operation.
No	GO to 16 .

I4 CHECK THE HVAC MODULE

NOTICE: The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.

- Start the engine.
- Connect the 12-volt incandescent test light between rear window defrost relay socket pin 2, circuit CRD02 (BN/BU) and rear window defrost relay socket pin 1, circuit CBB54 (VT/OG).



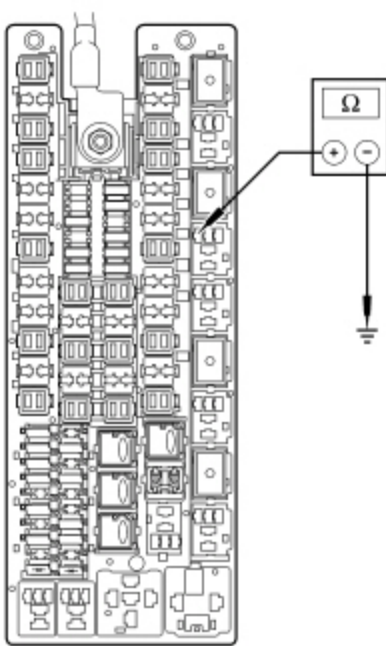
- N0101517

Does the test light illuminate?

Yes	GO to 15 .
No	INSTALL a new rear window defrost relay. TEST the system for normal operation.

I5 CHECK THE REAR WINDOW DEFROST RELAY CONTROL CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Disconnect: HVAC Module — EATC [C228A](#) or HVAC Module — Electronic Manual Temperature Control (EMTC) [C294A](#).
- Measure the resistance between rear window defrost relay socket pin 2, circuit CRD02 (BN/BU), harness side and ground.



• N0093252

Is the resistance greater than 10,000 ohms?

Yes	GO to 16 .
No	REPAIR short to ground in circuit CH122 (WH/OG). TEST the system for normal operation.

I6 CHECK FOR CORRECT HVAC MODULE OPERATION

- Disconnect all HVAC module connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all HVAC module connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new HVAC module. REFER to Section 412-01. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. TEST the system for normal operation.

Pinpoint Test J: The Delayed Accessory is Inoperative

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The accessory delay relay is located in the Body Control Module (BCM) . When the ignition is turned ON, the BCM activates the accessory delay relay by grounding the relay coil. When the ignition is turned to the OFF position, the BCM continues to ground the accessory delay relay coil for approximately 10 minutes, or until a front door is opened. When the accessory delay relay is active, it supplies voltage to the power window system.

- DTC B131A:12 (Delayed Accessory Power: Circuit Short To Battery) — sets when the BCM senses a short to voltage on the accessory delay relay control circuit.
- DTC B131A:14 (Delayed Accessory Power: Circuit Short To Ground or Open) — sets when the BCM senses an open or short to ground on the accessory delay relay control circuit.

This pinpoint test is intended to diagnose the following:

- Door ajar switches
- Accessory delay relay
- Wiring, terminals or connectors
- BCM

PINPOINT TEST J : THE DELAYED ACCESSORY IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

J1 CHECK FOR CORRECT OPERATION OF THE DOOR AJAR SWITCHES

- Open and close the front doors. Verify the interior lights turn on when the doors are open and off when the doors are closed.

Do the interior lights operate normally?

Yes	GO to J2 .
No	REFER to Section 417-02 to diagnose the interior lighting and door ajar switches.

J2 CHECK THE ACCESSORY DELAY RELAY

- Disconnect: Accessory Delay Relay .
- Carry out the accessory delay relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

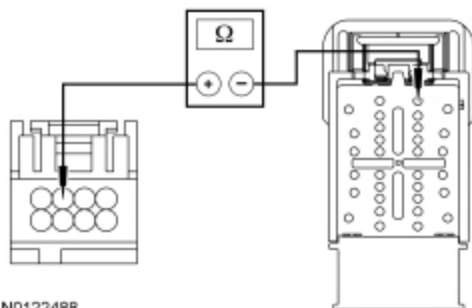
Did the delayed accessory relay pass the component test?

Yes	VERIFY Body Control Module (BCM) fuse 32 (15A) (regular cab) or BCM circuit breaker 48 (30A) (SuperCab or SuperCrew) is OK. If OK, GO to J3 . If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR the circuit. TEST the system for normal operation.
No	INSTALL a new accessory delay relay. TEST the system for normal operation.

J3 CHECK THE LH FRONT WINDOW CONTROL SWITCH POWER CIRCUIT FOR AN OPEN

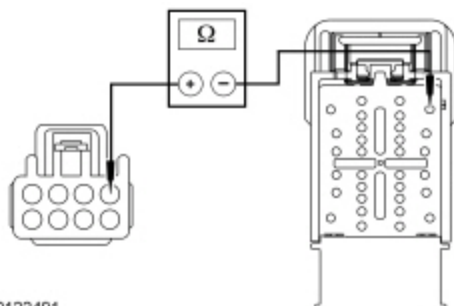
- Ignition OFF.
- Disconnect: BCM [C2280D](#) .
- Disconnect: LH Front Window Control Switch [C504](#) (Regular Cab) or [C535B](#)(SuperCab or SuperCrew) .

- For regular cab, measure the resistance between LH front window control switch [C504](#) Pin 3, circuit CBP32 (GN/VT), harness side and BCM [C2280D](#) Pin 17, circuit CBP32 (GN/VT), harness side.



• N0122488

- For SuperCab or SuperCrew, measure the resistance between LH front window control switch [C535B](#) Pin 1, circuit CBP01 (BU), harness side and BCM [C2280D](#) Pin 27, circuit CBP01 (BU), harness side.



• N0122491

Is the resistance less than 5 ohms?

Yes	GO to J4 .
No	REPAIR the circuit. TEST the system for normal operation.

J4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all BCM connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test K: The Delayed Accessory Does Not Turn Off

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The accessory delay relay is located in the Body Control Module (BCM) . When the ignition is turned ON, the BCM activates the accessory delay relay by grounding the relay coil. When the ignition is turned to the OFF position, the BCM continues to ground the accessory delay relay coil for approximately 10 minutes, or until a front door is opened. When the accessory delay relay is active, it supplies voltage to the power window system.

- DTC B131A:14 (Delayed Accessory Power: Circuit Short To Ground or Open) — sets when the BCM senses an open or short to ground on the accessory delay relay control circuit.

This pinpoint test is intended to diagnose the following:

- Accessory delay relay
- Wiring, terminals or connections
- BCM

PINPOINT TEST K : THE DELAYED ACCESSORY DOES NOT TURN OFF

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

K1 CHECK FOR STUCK/STICKING ACCESSORY DELAY RELAY

- Disconnect: Accessory Delay Relay .
- Ignition ON.
- Attempt to operate the power windows.

Do the power windows still operate?

Yes	GO to K3 .
No	GO to K2 .

K2 CHECK THE ACCESSORY DELAY RELAY

- Ignition OFF.
- Carry out the accessory delay relay component test.
Refer to Wiring Diagrams Cell [149](#) for component testing.

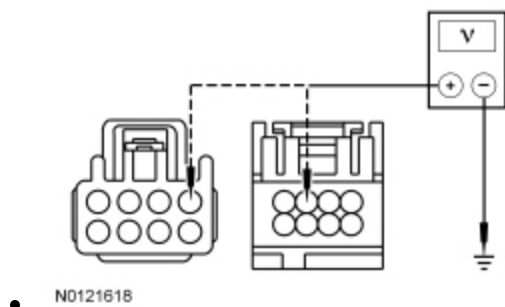
Did the accessory delay relay pass the component test?

Yes	GO to K4 .
No	INSTALL a new accessory delay relay. TEST the system for normal operation.

K3 CHECK THE LH FRONT WINDOW CONTROL SWITCH POWER CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.

- Disconnect: BCM [C2280D](#) .
- Disconnect: LH Front Window Control Switch [C504](#) (Regular Cab) or [C535A](#) (SuperCab or SuperCrew) .
- Ignition ON.
- Measure the voltage between LH front window control switch and ground.
 - For regular cab, measure the voltage between [C504](#) Pin 3, circuit CBP32 (GN/VT), harness side and ground.
 - For SuperCab or SuperCrew, measure the voltage between [C535B](#) Pin 1, circuit CBP01 (BU), harness side and ground.



Is any voltage present?

Yes	REPAIR circuit CBP01 (BU) or CBP32 (GN/VT) for a short to voltage. TEST the system for normal operation.
No	GO to K4 .

K4 CHECK FOR CORRECT BCM OPERATION

- Disconnect all BCM connectors.
- Check for:
 - corrosion.
 - pushed-out pins.
- Connect all BCM connectors and make sure they seat correctly.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new BCM. Refer to the appropriate Removal and Installation procedure in Section 419-10. . TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test L: All Passenger Power Windows are Inoperative — LH and RH Rear

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

Normal Operation

The LH and RH rear window control switches receive voltage from the LH front window control switch when the lock-out switch is in the UNLOCK position. Ground is also provided to the LH and RH rear window control switches from the LH front window control switch. The LH and RH rear window control switches transfer voltage and ground to the power window motor. Window direction is controlled by reversing polarity of the voltage and ground circuits being supplied to the power window motor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- LH front window control switch

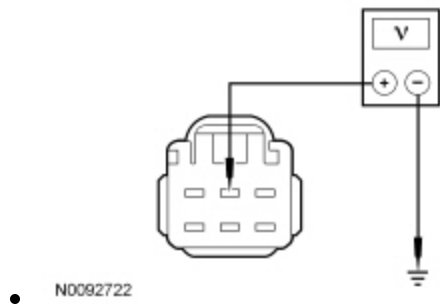
PINPOINT TEST L : ALL PASSENGER POWER WINDOWS ARE INOPERATIVE — LH AND RH REAR

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

L1 CHECK THE VOLTAGE TO THE LH REAR WINDOW CONTROL SWITCH

- Disconnect: LH Rear Window Control Switch [C701](#).
- Ignition ON.
- **NOTE:** Set the LH front window lock-out switch to the UNLOCK position.

Measure the voltage between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and ground.

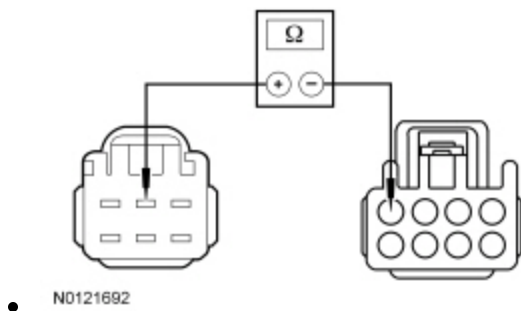


Is the voltage greater than 10 volts?

Yes	GO to L3 .
No	GO to L2 .

L2 CHECK THE WINDOW CONTROL SWITCH POWER CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between LH rear window control switch [C701](#) Pin 2, circuit CPW14 (VT/WH), harness side and LH front window control switch [C535B](#) Pin 4, circuit CPW14 (VT/WH), harness side.

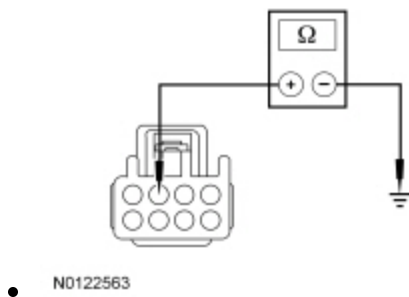


Is the resistance less than 5 ohms?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

L3 CHECK THE LH FRONT WINDOW CONTROL SWITCH GROUND CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: LH Front Window Control Switch [C535B](#).
- Measure the resistance between LH front window control switch [C535B](#) Pin 3, circuit GD134 (BK/VT), harness side and ground.



Is the resistance less than 5 ohms?

Yes	INSTALL a new LH front window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
No	REPAIR the circuit. TEST the system for normal operation.

Pinpoint Test M: The One-Touch Up/Down Feature is Inoperative

Refer to Wiring Diagrams Cell [100](#) , Power Windows for schematic and connector information.

NOTE: A new power window motor will not operate in one-touch up mode until initialized. Refer to Window Motor Initialization in this section.

Normal Operation

During normal operation, when the window control switch is pressed to the second detent position to request a one-touch up or down operation, the window control switch provides a 12-volt signal on both the up and down line simultaneously. During a one-touch up or down request, the window motor determines intended window direction by the signal first received (first detent position). If the window control switch is pressed too quickly to the second detent position (less than 5 milliseconds of time between first detent signal and second detent signal), the window motor will not be able to determine the intended direction request and will not operate until the window control switch is released and pressed again.

This pinpoint test is intended to diagnose the following:

- Power window motor not initialized
- Window control switch
- Window regulator motor

PINPOINT TEST M : THE ONE-TOUCH UP/DOWN FEATURE IS INOPERATIVE

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

M1 CHECK FOR POWER (B+) LOSS DURING WINDOW OPERATION

- Verify if power (B+) was lost during window operation.

Was power (B+) lost while the window was in operation?

Yes	RAISE the suspect window to the fully CLOSED position. VERIFY the window glass stalls into the upper header seal. The window should function normally. TEST the system for normal operation.
No	GO to M2 .

M2 DE-INITIALIZE, THEN INITIALIZE THE WINDOW MOTOR

- De-initialize the suspect window motor, then carry out the window motor initialization procedure. Refer to Window Motor Initialization in this section.

Is the one-touch down feature operating correctly?

Yes	The system is operating normally at this time. The window motor lost initialization.
No	GO to M3 .

M3 CHECK THE SUSPECT WINDOW CONTROL SWITCH

- Ignition OFF.
- Disconnect: Suspect Window Control Switch C504 (LH front, Regular Cab) C535A, C535B (Supercrew or SuperCab), or C604 (RH front) .
- Carry out the component test for the suspect window control switch. Refer to Wiring Diagrams Cell [149](#) for component testing.

Did the window control switch pass the component test?

Yes	INSTALL a new window regulator and motor. REFER to Window Regulator and Motor — Front Door in this section. INITIALIZE the window motor. REFER to Window Motor Initialization in this section. TEST the system for normal operation.
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No	INSTALL a new window control switch. REFER to Window Control Switch — Front in this section. TEST the system for normal operation.
-----------	--

Component Test

Grid Wire Test

1. Using a bright lamp in the vehicle, inspect the wire grid from the exterior. A broken grid wire will appear as a brown spot.
2. Run the engine at idle. Set the rear window defrost switch to ON. The indicator light should come on.
3. Working in the vehicle with a voltmeter, contact the broad red-brown stripes of the rear glass window positive lead to battery side and the negative lead to ground side. The meter should read 10-13 volts. A lower voltage reading indicates a loose ground connection.
4. Contact a good ground point with the negative lead of the meter. The voltage reading should not change.
5. With the negative lead of the meter grounded, touch each grid line of the heated rear window glass at its midpoint with the positive lead. A reading of approximately 6 volts indicates the line is good. A reading of 0 volt indicates the line is broken between the midpoint and the B+ side of the grid line. A reading of 12 volts indicates the circuit is broken between the midpoint of the grid line and ground.
6. Pinpointing the exact position of the break can be accomplished (if the voltmeter reads 0 volt when the midpoint of the grid line is touched with the positive lead of the voltmeter) by moving the positive lead of the voltmeter toward the B+ side of the grid line and touching the grid line until the voltmeter reads 12 volts. If the voltmeter reads 12 volts when the midpoint of the grid line is touched with the positive lead of the voltmeter, simply move the positive lead of the voltmeter toward the ground connection of the grid line and touch the grid line until the voltmeter reads 0 volt.